

EDUCATION

B.E. in Electronics and Communication Engineering — SSN College of Engineering, Chennai

May 2024

GPA: 8.6/10

WORK EXPERIENCE

Software Engineer — Visteon Technical and Services Pvt Ltd

Jan 2024 – Present

- Architected containerized Java 21 / Spring Boot microservices for the CAPL generation framework — parsing Excel and .cdd diagnostic files via Apache POI and DOM-based XML traversal to auto-produce UDS test scripts, generating both CAPL and Robot Framework (.robot) test suites covering session sequencing, security handshake, DID definitions, and negative response handling — preserving ~80% reduction in manual authoring time.
- Built the reporting web app as a full-stack platform — a React (Vite) frontend and 3 independently deployable Spring Boot REST microservices behind an API gateway, powering existing Chart.js dashboards and Excel/PDF export — containerized and orchestrated via Docker Compose for cloud-ready deployment.
- Designed and owned an end-to-end Bluetooth and Wi-Fi connectivity automation framework in Python — integrated NI VBAI (Vision Builder for Automated Inspection) for image-based cluster display capture via Vision Acquisition, authored a fuzzy text-matching engine in DataExchange.py to reliably identify device names under varying render conditions, orchestrated PRB joystick navigation over serial, and performed ADB passkey validation; reduced manual connectivity test time by 95% across multiple OEM programs.
- Architected and shipped a native Kotlin Android application using Google ML Kit On-Device OCR to capture cluster display passkeys in real time via the camera, piped the extracted text into a Python automation pipeline through ADB logcat, and eliminated the last manual touchpoint in the BT pairing flow — enabling fully zero-touch autonomous pairing end to end.
- Built a production-grade TPMS BLE sensor monitoring Android app in Kotlin — designed the full application architecture with MVVM (ViewModel + LiveData/StateFlow), implemented low-level Eddystone-URL/TLM advertisement packet parsing to decode live tyre pressure and temperature payloads, engineered OEM-specific BLE scan filter configurations and connection workarounds for reliable multi-sensor discovery across all four wheels.
- Architected and deployed **V-RoboBus**, a production-grade **React + REST API** CAN automation platform using **worker threads** and **WebSockets** for non-blocking multi-channel CAN/CAN-FD tracing, signal simulation, live updates, and UDS diagnostics. Integrated with **Robot Framework** and rolled out for **Visteon-wide use**, supporting Vector/PCAN hardware and DBC/CDD-driven validation workflows.
- Refactored core Python automation modules by applying efficient data structures and algorithmic optimizations — profiled bottlenecks, restructured data pipelines, and improved overall framework execution speed, reducing defect escape rate by 85%.

RELEVANT PROJECTS

ScoreStep — Credit Health Coach Android App (Kotlin + Jetpack Compose)

Jun 2026

- Architected a production-grade offline-first Android app in Kotlin using MVVM (ViewModel + StateFlow), Jetpack Compose, Room DB, Hilt DI, and Jetpack Navigation — designed the full application architecture from scratch with a clean Repository pattern and single source of truth.
- Built a custom Canvas-drawn animated DTI (Debt-to-Income) gauge in Compose with smooth arc animation and colour-coded eligibility scoring (green/amber/red), alongside a real-time what-if simulator with income and EMI sliders that recalculates loan eligibility instantly on-device.
- Implemented a personalised action plan engine that auto-generates week-by-week credit improvement steps from the user's financial profile — covering EMI reduction, CIBIL report review, credit utilisation, and buffer fund strategies — with persistent completion tracking via Room, plus a 3D animated splash screen using Compose Canvas.

LoanLens — Fintech Loan Risk Analyzer (Java Spring Boot + React)

Dec 2025 – Mar 2026

- Engineered a production-grade full-stack fintech application with 9 secured REST APIs — JWT auth, loan CRUD, stress simulation, portfolio aggregation, affordability scoring, and iText PDF report generation using Java Spring Boot 3 and MySQL.
- Built React 18 frontend with Redux Toolkit, Chart.js health gauge, DTI progress bar, amortization table, and real-time EMI calculator using the reducing-balance formula with BigDecimal precision.
- Containerized with multi-stage Docker builds and deployed via Kubernetes (HPA 2–6 pods) on AWS EKS with GitHub Actions CI/CD (lint, test, Trivy scan, blue-green deploys); covered with JUnit 5 + Mockito, Testcontainers, Playwright E2E, and k6 load tests (p95 < 800ms @ 100 concurrent users).

Best & Banting — Full-Stack Medical Equipment E-Commerce Platform (React, Node.js, PostgreSQL)

Dec 2025

- Architected a multi-page React 18 SPA (React Router) with a data-driven product catalog — dynamic category/type/detail routes generated from a single taxonomy source — and an interactive 3D product viewer built with React Three Fiber/Three.js (GLB models, orbit controls) with GSAP ScrollTrigger-driven camera animation.
- Engineered a secure Node.js/Express REST API with JWT auth, bcrypt password hashing, and a custom in-memory rate limiter for brute-force login protection; integrated Supabase (PostgreSQL) with parameterized queries, server-side filtering/pagination, and role-restricted admin endpoints.
- Built an admin analytics dashboard from scratch with no charting library (custom SVG/CSS donut/bar charts, monthly trend visualizations, CSV/HTML export), Resend API transactional email on enquiry submission, and deployed a split Netlify + Render architecture with production CORS, env-based secrets, and a self-pinging keep-alive job.

Fleet Management Solution with RBAC in C++

Jun 2025

- Secured user authentication via SHA256 password hashes, processed rental/return workflows with real-time cost computation, and generated admin reports using structured SQL queries with date-range filters, with an interactive HTML/CSS/JS UI validated against 500+ login attempts and DB-level audit tracking.

Virtual Automotive Control System — Best Team Award, Gear Up Hackathon (Visteon & KICIT, 100+ teams)

Mar 2025

- Built a complete C automotive system simulating Engine Control, Transmission, Braking, HVAC, Body Control, and Instrument Cluster; awarded Best Team by Yashavant Kanetkar and senior Visteon engineers.

Retinal Disease Diagnosis using Deep Learning

May 2024

- Trained a Multi Scale-Attention network on fundus and OCT images achieving ~93% accuracy for multi-class retinal disease classification; deployed via Django web app with real-time diagnosis output.

TECHNICAL SKILLS

Languages: C, C++, Python, Java, JavaScript, Kotlin, SQL, .NET, Robot Framework**Web & Frontend:** React.js, Redux, Angular.js, Node.js, Express.js, Flask, FastAPI, Django, REST APIs, HTML, CSS3, Chart.js, SheetJS**Android & Mobile:** Kotlin, Android SDK, Jetpack Compose, XML Views, ViewModel, LiveData/StateFlow, Room, Google ML Kit, BLE/Eddystone, ADB, Appium**Databases:** MySQL, MongoDB, SQLite, Redis**Automotive & Embedded:** CANoe, CANoe.DiVa, CAPL, UDS/CAN Diagnostics, PySide6/Qt, python-can, cantools, NI VBAI, Vision Acquisition, Windows BCrypt API, ADB, Appium, Google ML Kit**Security:** AES-128-ECB, SHA256, JWT, BCrypt, SMTP Authentication, OWASP**Cloud, DevOps & Tools:** AWS, Docker, Kubernetes, GitHub Actions, Jenkins, Terraform, JIRA, Postman, JUnit, Playwright